

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

7 Data Science (djw1005)

We are given a numerical dataset $\{x_1, x_2, \dots, x_n\}$. We wish to estimate the 99th percentile, and to find a confidence interval for it. Here are three approaches:

(a) We may decide to model the datapoints as independent samples from the $\text{Pareto}(1, \alpha)$ distribution. Then, the 99th percentile is the value q such that $\mathbb{P}(\text{Pareto}(1, \alpha) \leq q) = 0.99$.

(i) Find the maximum likelihood estimator for α . [3 marks]

(ii) Find q as a function of α . [2 marks]

(iii) Explain how to use parametric resampling to find a confidence interval for q . Give pseudocode. [4 marks]

(b) We may decide to estimate the 99th percentile by simply sorting the dataset and reading off the value in position $\text{int}(0.99n)$.

Explain how to use nonparametric resampling to find a confidence interval for it. Give pseudocode. Under what circumstances would you expect the result to be unreliable? [6 marks]

(c) We may decide to use computational Bayesian methods to find the confidence interval. Explain how, stating your model precisely. Give pseudocode. [5 marks]

Hint. If $X \sim \text{Pareto}(1, \alpha)$ then it has cumulative distribution function

$$\mathbb{P}(X \leq x) = \begin{cases} 1 - x^{-\alpha} & \text{if } x \geq 1 \\ 0 & \text{if } x < 1. \end{cases}$$

Answer:

— *Solution notes* —

Part (a)(i). The pdf for $\text{Pareto}(1, \alpha)$ is

$$\Pr(x) = \alpha x^{-\alpha-1}, \quad x \geq 1$$

so the log likelihood for the entire dataset (assuming all datapoints are ≥ 1) is

$$\log \Pr(x_1, \dots, x_n) = n \log \alpha - (\alpha + 1) \sum \log x_i.$$

The mle is the α that maximizes this. Setting the derivative equal to zero,

$$\frac{n}{\hat{\alpha}} - \hat{\alpha} \sum \log x_i = 0 \quad \Rightarrow \quad \hat{\alpha} = \frac{n}{\sum \log x_i}.$$

Part (a)(ii). Using the given cdf, and letting $p = 0.99$,

$$1 - q^{-\alpha} = p \quad \Rightarrow \quad q = (1 - p)^{-1/\alpha}.$$

Part (a)(iii). Define a function that computes $q(\hat{\alpha}(\mathbf{x}^*))$ for an arbitrary dataset $\mathbf{x}^*$. Define a function X^* that generates n independent samples from $\text{Pareto}(1, \hat{\alpha}(\mathbf{x}))$ where $\mathbf{x}$ is the dataset we're given. Generate a large number of samples of $q(\hat{\alpha}(X^*))$ and find say a 95% confidence interval with `np.quantile(samples, [.025, .975])`. *[Any pseudocode that expresses the above is acceptable. The point of asking for pseudocode is to encourage students to think through every part of what's needed in this procedure.]*

Part (b). Define a function $q(\mathbf{x}^*)$ that returns the empirical 99th percentile, as given in the question. Define a function X^* that generates a random dataset of size n by sampling with replacement from the dataset we're given. Generate a large number of samples of $q(X^*)$ and compute a confidence interval as above.

The fundamental 'wishful thinking' behind resampling is that resampled datasets should adequately represent future datasets that we might see. When we're asking for an extreme statistic such as $q(X^*)$, we're assuming that this is representative of future datasets. If there are very 'sparse' datapoints in this part of the distribution, e.g. if $0.01n$ is small, then this resampling is likely to be unreliable. *[This asks for thinking that goes beyond pure bookwork. 1 mark for just saying "small n ". 2 marks for something more.]*

Part (c). First decide on a prior distribution for α . Generate a large sample of values from this prior. Then, for each sampled α_i , let w_i be the likelihood of the dataset assuming parameter α_i , as calculated in part (a)(i). Then rescale the w_i to sum to 1. Then find a range containing 95% of the w_i mass, using the standard sort-then-cumulative-sum trick from lecture notes.
